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# Numerical simulation and structural recombination of microchannel micromixer for excellent mixing performance

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ABSTRACT

A facile continuous flow micromixer with a novel internal structure is developed to improve the mixing effect through computational fluid dynamics (CFD) simulation. Meanwhile, an authentic transparent micromixer was prepared following the mathematical model of the micromixer, and reliability of the simulation was verified by result comparison of fluid mixing performance. Moreover, several internal structures were designed and their mixing effect was calculated. Using velocity inlet and pressure outlet, ignoring the change of energy during the mixing process, it is shown that the structure of the micro-mixer has a positive effect on the mixing effect at $Re <$ 1200. In the cubic structure, the fluid is dominated by molecular diffusion, the increase of concavity and convexity structure makes the fluid produce spiral flow, and the fluid separates and recombines many times in the mixed structure, the convective mixing process is superior to the molecular diffusion process and the mixing effect is improved. Furthermore, it was astonished that the addition of cubic structure inside convex-concave structures (a recombination structure), improve the mixing index as well as reducing the flow resistance by increasing fluid-to-fluid interface area.

## 1. Introduction

Microfluidic technology has attracted much attention for its uniform and rapid mixing of fluids in microchannels [1]. In the micro-reaction, liquids are mixed through unique micro-channels to enhance the mass transfer and heat transfer [2–4]. Compared with the traditional batch micromixer or tank micromixer, the microchannel micromixers enable a rapid mixing of fluids; the continuous operation leads to high production efficiency, as well as increased safety; micromixers merely require energy from the pump to mix the reactant.

After decades of technological development, micro-reaction has been widely applied in industrial production, for example, in the field of inorganic chemicals [5], organic chemicals [6], biology [7], medicine [8], and bio-pharmaceuticals [9]. The efficiency of chemical reactions is determined by mass transfer and mixing between two phases. Therefore, the principles of mass transfer, heat transfer, and flow regime in microchannels have been extensively studied [10]. According to the discrepancy of the reaction system, the micro-channels are equipped with deformable baffles to obtain high mixing index [11]. Depending on the mixing structures, microchannel micromixers can be classified into T-shaped mixers [1], split and recombine mixers [12], Jet mixers [13], etc. Among them, the T-shaped mixers allow violent mixing that collides at the corner, with the more effective mixing through the extended outlet pipe. The split and recombine is desirable for multiple separations and combination to strengthen mixing performance. In the Jet mixers, by changing the incident angle, blocking for chemical reactions comprising solid materials can be successfully avoided.

Numerical simulation is one of the essential methods to evaluate the mixing effect of a preliminary designed micromixer [14]. Through CFD, the flow regime and mixing impact can be determined, and the results of CFD can then be used to optimize the structural model. [15]. Hermann et al. reconstructed the split and recombine micromixer based on the numerical simulation and proposed the structural model with optimal mixing effect was obtained at $Re \leq 700$ [12]. Hossain et al. [16] pointed out that the micromixer with zig-zag structure displayed the best mixing performance (at $0.267 \leq Re \leq 267$) among of a zig-zag, square-wave, and

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**Table 1**
The devices of Passive mixing.

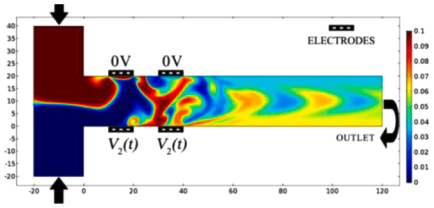
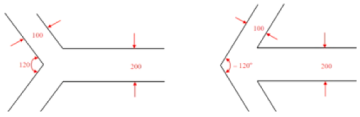
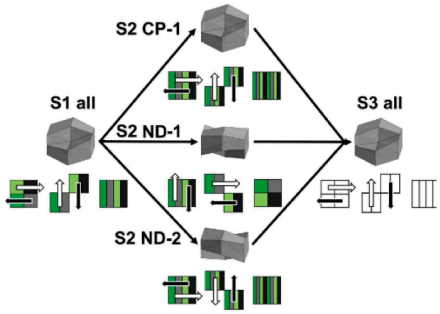
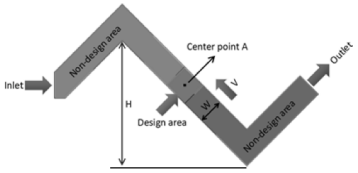
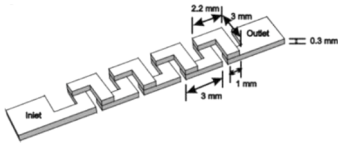
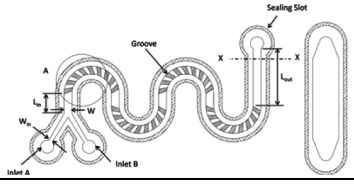
| Type | | Application | Schematic diagram | Source |
|---|---|---|---|---|
| T-type | | Gas-liquid,liquid- liquid |  | Cheng et al. (2018) [39] |
| Y-type | | Gas-liquid,liquid- liquid |  | Hsieh et al. (2013) [40] |
| Split and Recombine | | Gas- liquid,liquid- liquid |  | Schönfeld et al. (2004) [41] |
| Others | Z-type | Gas- liquid,liquid- liquid |  | Chen X et al. (2016) [42] |
| | L-type | Gas- liquid,liquid- liquid |  | Vijayendran R A et al. (2003) [43] |
| | arc-type | Gas- liquid,liquid- liquid |  | Kristina J. Cook et al. (2013) [44] |

**Table 2**
Physical properties of fluids.

| Fluid | Density (kg $m^{-3}$) | Viscosity (Pa s) |
|---|---|---|
| A (pure water) | 997 | $0.89 \times 10^{-3}$ |
| B (NaCl aqueous solution) | 1072 | $1.08 \times 10^{-3}$ |

curved forms. Zheng *et al.* reported the mixing result of two-phase gas/liquid flow in a microchannel with complex system through numerical simulation and experimental data and further optimized the baffle angle to 67.5°for a better mixing effect [17]. Mondal *et al.* presented that the raccoon and serpentine micromixers showed a relatively similar mixing effect ($0.1 < Re < Re^*$) through numerical simulation, but the Raccoon micromixer exhibited better mixing effects at $Re > Re^*$ [18]. Hsieh *et al.* optimized the angle of the Y-type micromixer to −60° for the better mixing effect [19]. Soleymani et al. developed the operating parameters of T-shaped micromixers, and found that the occurrence of eddy currents was beneficial for fluid mixing [20]. Mashaei *et al.* designed an improved curved micromixer with increased mixing index by 2.04 times compared with a simple curved micromixer. However, the pressure drop only increased by 1.23 times [21]. Moreover, different micromixers, such as H-shaped [22], L-shaped [23], and Y-shaped [19], were designed to investigate their flow further and mixing properties. Volker Hessel et al. [34] given a review on micro-structured mixer devices. In this paper, the mixing principles applied can be divided in two classes relying either on the pumping energy or provision of other external energy to achieve mixing, termed passive and active mixing, respectively. And the mixing principle, design elements and mixing efficiency analysis are summarized. Chia-Yen Lee et al. [35] summarized the effects of different channel types on mixing in passive mixing.

To enhance the mixing effect in the microchannel, the optimization of mixing structures and length can also be applied here [24]. Shah *et al.* studied mixing units in Y-shaped split and recombine micromixers. The mixing index of the micromixers with circular and rhombus was 0.99 at $Re = 100$, which is higher than other proposed micromixers [25]. Compared to the micromixers with identical mixing units, recombining different mixing units also enhances the mixing effect [12]. Simultaneously, obstacles of the mixing structure, such as baffle [26], semi-circular [27] and, groove [28], can change the fluid path and ultimately

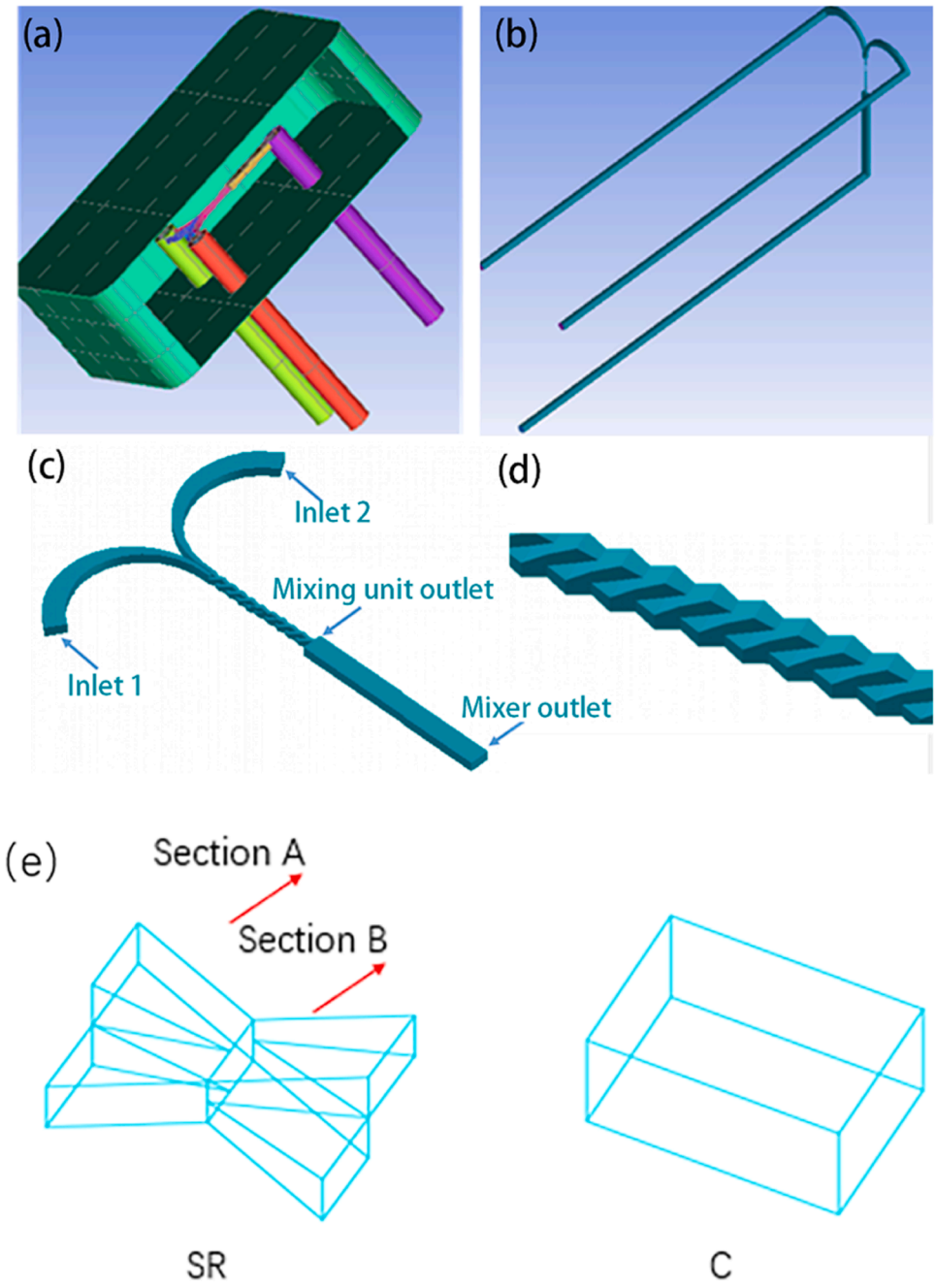



**Fig. 1.** Model diagram of micromixer. (a) the micromixer (b) Internal flow road (c) Mixed section (d) split and recombine mixing unit outlet (e) SR type micromixer and C type micromixer.

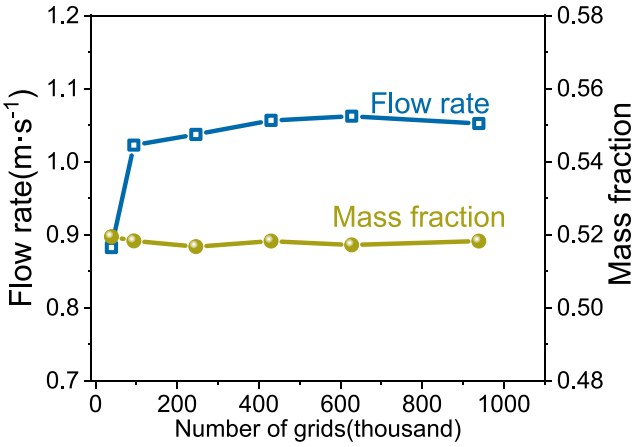



**Fig. 2.** Variation of flow rate and mass fraction in center points of outlet.

**Table 3**
Geometric parameters.

| Name | Geometric parameters |
|---|---|
| Mixing length | 0.6cm |
| Number of mixed units | 10 |
| Height of concave section | 0.25—0.5cm |
| Inlet flow | 5—30ml·min$^{-1}$ |

improve the mixing performance [27,29]. Hossain *et al.* [30] proposed the micromixer with crossing channels, which display the excellent mixing effect at the *Re* range from 0.2 to 120, the mixing index was improved by increasing the contact area of the fluids and optimizing mixing structures. Mohamad Ali Aghasi et al. [36] designed six novel micromixers what are based on the splitting and recombination mechanism and investigated. In the range of *Re* from 0 to 100, the mixing index (MI) of a micromixer with seven alternating mixing units is higher than 0.92. The mixing index was improved by changing the number of cells and the direction of flow. Zahra Far et al. [37] in order to model the cylinder the vibration, utilized the mass-spring-damper system. The effect of vortex-induced vibration of an elastically mounted rigid circular cylinder with two degrees of freedom movement on the mixing process in a straight channel micromixer is studied. By changing the deceleration speed and Peclet number of the fluid, it is found that the VIV can improve the mixing performance under the condition of high Peclet number. From the above discussion, the forms of combining units have a significant influence on the mixing effect, to allow the application of the micromixers in the broader range of flow rates. Nguyen et al. [38] discusses the operation points of the micromixers based on characteristic dimensionless numbers such as Reynolds number *Re*, Peclet number Pe, and in dynamic cases the Strouhal number St. And the type, application and schematic diagram of Passive mixing were summarized in Table 1.

In the present work, we studied the mixing flow characters of split and recombine and cube micromixers by CFD simulation. And a mock up is made to verify the mixing effect of split and recombine micromixers.

**Table 4**
Simulate Reynolds number and flow value range.

| Number | Reynolds number | Flow rate (mL min$^{-1}$) |
|---|---|---|
| 1 | 22.15 | 1 |
| 2 | 66.46 | 3 |
| 3 | 110.76 | 5 |
| 4 | 221.52 | 10 |
| 5 | 443.04 | 20 |
| 6 | 664.56 | 30 |
| 7 | 886.08 | 40 |
| 8 | 1107.60 | 50 |

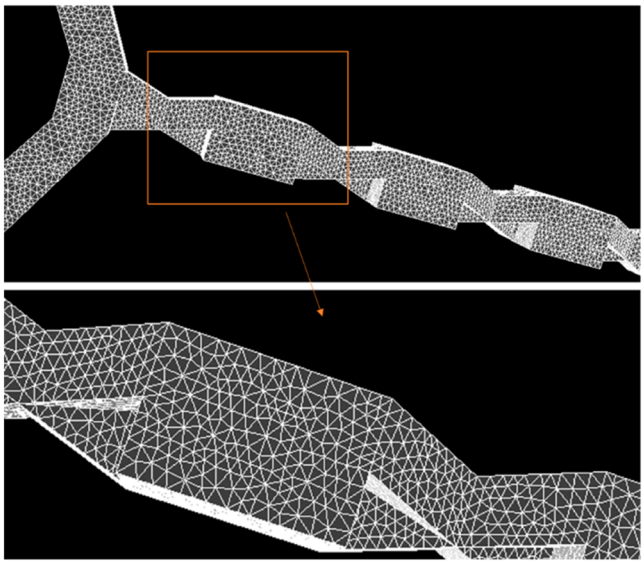


**Fig. 3.** A typical grid distribution for SR.

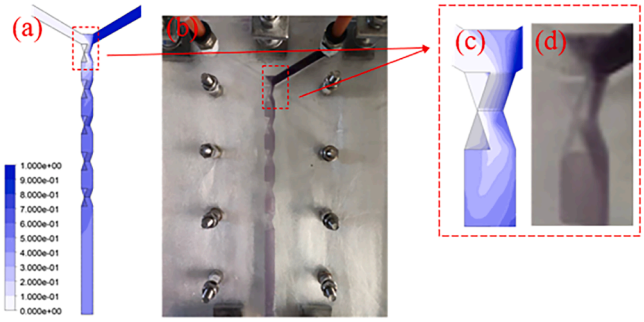



**Fig. 4.** Comparison between experiment and simulated mixing process. (a) Simulation Diagram (b) Experimental drawing (c) (d) Partially enlarged view.

Then, the structure size of split and recombine section height was optimized through quantitative calculation of concentration distribution, flow rate and mixing index. Furthermore, the X type split and recombine micromixer, various structures combined with the cubic units would be calculated for better performance. The results will guide the design of microchannel mixer with high efficiency and low flow resistance.

## 2. Numerical simulations

### 2.1. Governing equation

The simulations were performed under steady-state conditions in Ansys Fluent 19.2. The continuity equation (Eq. (1)) and momentum equation (Eq. (2)) were used to calculating the flow in the micromixer:

$$\nabla \cdot U = 0 \tag{1}$$

$$\rho \frac{\partial U}{\partial t} + \rho(U \cdot \nabla)U = -\nabla P + \mu \nabla^2 U \tag{2}$$

Convection Diffusion equation:

$$\frac{\partial C}{\partial t} + (U \cdot \nabla)C = D\nabla^2 C \tag{3}$$

He, $U$ is the fluid flow rate, m/s.
$\mu$ is the liquid viscosity, Pa·s.
$P$ was the pressure, Pa.
$C$ is the compound concentration, mol/L.

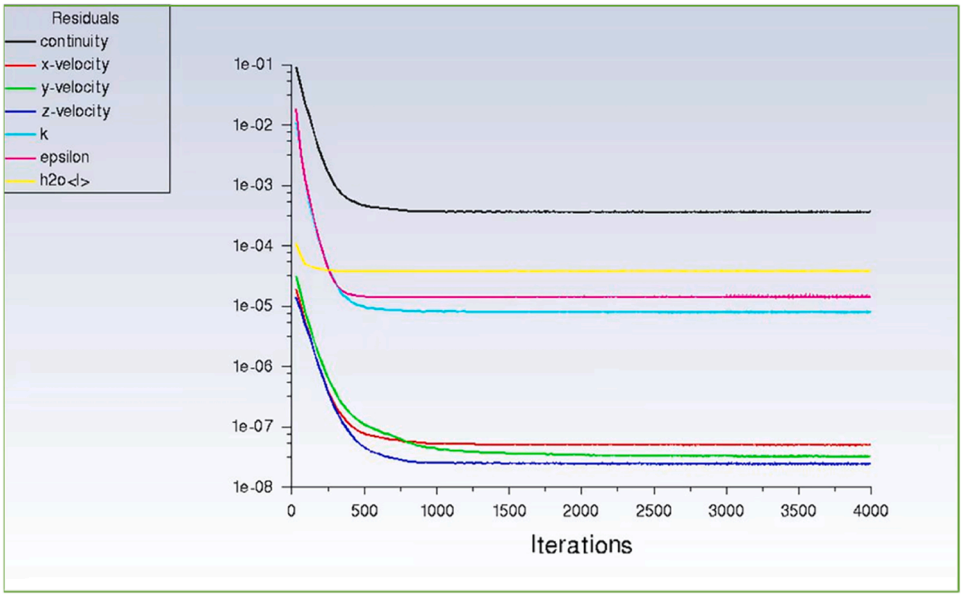



**Fig. 5.** Residuals simulation calculation chart.

$D$ is the component diffusion coefficient, $m^2/s$.

Besides, the Navier-Stokes equation is the most widely used model for calculation. It takes the time average for the brief turbulence fluctuation term. In this paper, considering the computational resources, convergence and violent pulsating flow state of fluid the Stand K-epsilon model of N-S model is selected. According to the above theoretical basis, the mixing of fluid A and B was simulated, and the physical properties are listed in Table 2. The diffusion coefficient between the two fluids was $1 \times 10^{-9}\ m^2\ s^{-1}$.

Herein, the mixing index is used to evaluate the mixing effect of the micromixer. Specifically, $\sigma^2$ is expressed by the variance of the material concentration of the cross-section of the micromixer, and the calculation method is as follows:

$$MI = 1 - \sqrt{\frac{\sigma^2}{\sigma^2_{max}}} \tag{4}$$

$$\sigma^2 = \frac{\sum (C_i - \overline{C}_m)^2}{N} \tag{5}$$

where $C_i$ represents the mass fraction of a node on the cross-section, $\overline{C}_m$ said the average mass fraction on the cross-section under ideal conditions, N is the number of nodes on the team, and $\sigma^2_{max}$ is the mass fraction with maximum variance. This paper uses the material concentration at different nodes to calculate the degree of liquid mixing in terms of MI.

### 2.2. Model design and meshing

#### 2.2.1. Model design

In this work, four kinds of mixers were modeled and calculated, concluding simple cube (C) micromixers, split and recombine (SR) micromixer, X-shaped split and recombine (XSR) micromixer, recombination X-shaped split and recombine (IXSR) micromixer.

The external shape of micromixer is a cube with an dimension of 10 cm × 6 cm × 5 cm, with two inlets (red and green) and one outlet (purple), as shown in Fig. 1(a). And the inlets and outlet were set to the exact dimensions as 1 mm × 0.5 mm for all kinds of micromixers and C micromixers. Fig. 1(b) and (c) flowed channel diagrams of the internal and critical sections in micromixer, respectively. The length of the microchannel is divided into 10 mixing units with 0.6 mm, as shown in Fig. 1(d). The mixing units of different structures are shown in Fig. 2(e). The dimensions of Section A remain unchanged, and the height of concave section B in split and recombine micromixer is 0.25, 0.375, and 0.5 mm (denoted as SR0.25, SR0.375 and SR0.5, respectively) with an aspect ratio of 1.25. The X-shaped split and recombine micromixer (XSR0.25) structure of was fully bonded, whereas SR0.25 was only half of that. The C type mixer is a simple cube, realizing the free flow of fluid inside it without being affected by the structure. These five micromixers were designed to explore the mixing effect. The geometric parameters are summarized in Table 3:

#### 2.2.2. Meshing

The model is divided into the unstructured mesh. The global dimensions of 0.006 cm, 0.007 cm, 0.008 cm, 0.01 cm, 0.014 cm and 0.02 cm correspond to the number of grids of 939,000, 628,000, 431,000, 246,000, 94,000 and 39,000, respectively, the independence of the grid is judged by monitoring the velocity and concentration of the exit center point.

### 2.3. Boundary and initial conditions

#### 2.3.1. Numerical methodology

The second-order upwind method is employed to discretize the convective terms. The standard second-order method is used to discretize the pressure. Convergence criteria is set $10^{-3}$ for all equations.

#### 2.3.2. Boundary conditions

No-slip boundary condition was set at all the solid surfaces.

#### 2.3.3. Inlet

The flow rate condition was used at the inlet boundary.

#### 2.3.4. Outlet

The pressure outlet condition was used at the outlet boundary.

The independence of the grid is judged by monitoring the flow rate and the mass fraction of mixed solution at the outlet center point. The two inlets were fluid A and B, respectively. Turbulence intensity and hydraulic diameter are relevant to inlets and outlets. Table 4 shows the inlet flow rates associated with *Re* at the range from 22.15 to 1107.60.

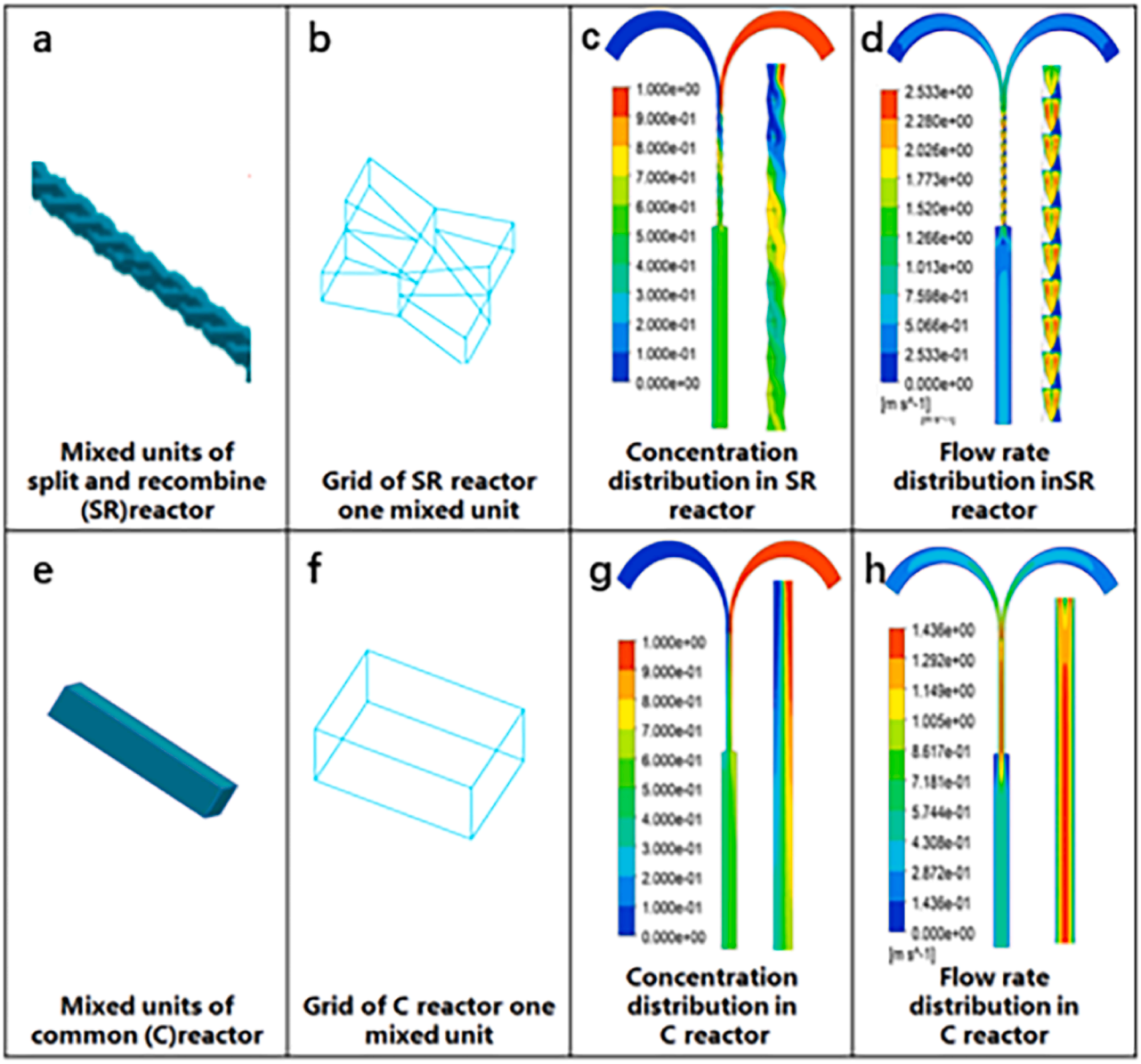



**Fig. 6.** Effect of micromixer structure on concentration field and flow rate field.

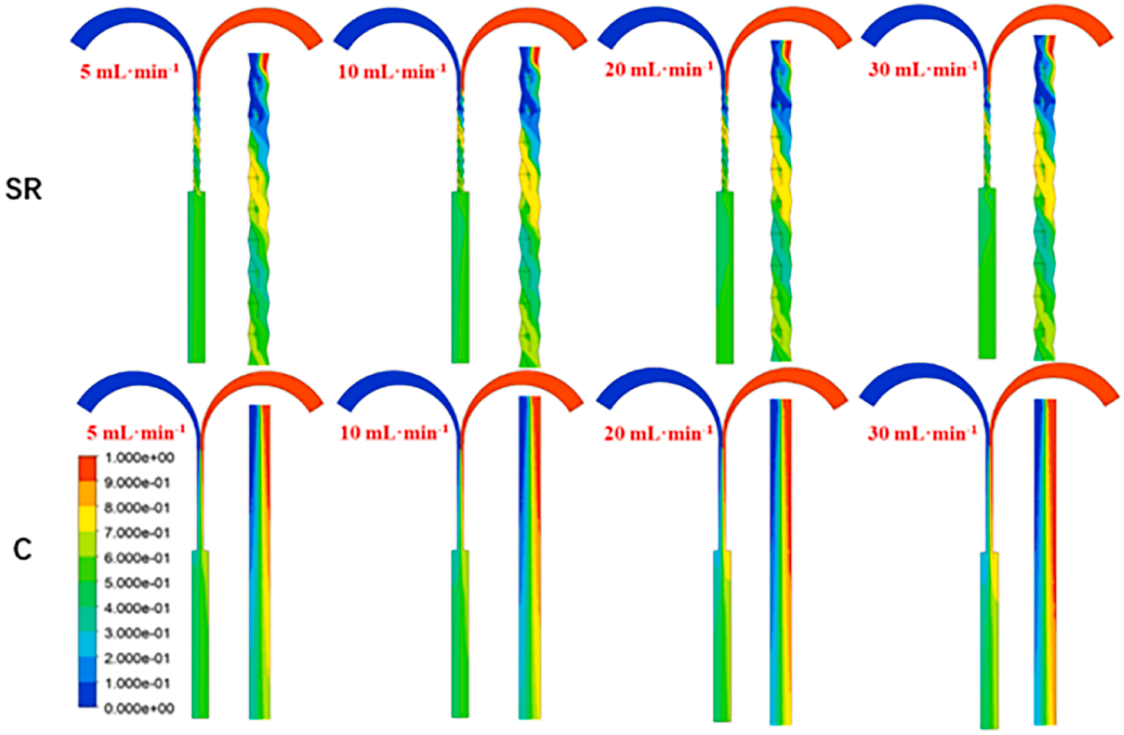



**Fig. 7.** Concentration distribution of SR micromixer and C micromixer reactors at different flow rates.

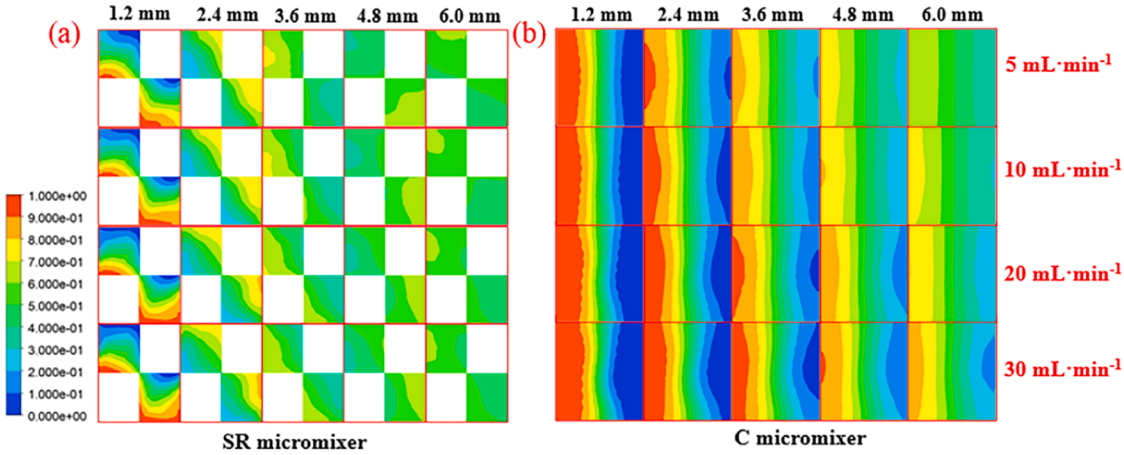



**Fig. 8.** Cross section concentration at different flow rates. (a) Concentration distribution of SR micromixer (b) Concentration distribution of C micromixer.

## 3. Results and discussion

### *3.1. grid independence and model validation*

As shown in Fig. 2, the solution flow rate and mass fractions of mixed solution at export have begun to stabilize when the number of grids is 431,000. However, we use 939,000 grids for the following calculation to ensure better calculation accuracy. To accomplish accurate numerical results in the current mixing study, a high-quality grid is used for all geometries Fig. 3

The mixing process of two fluids in the micromixer was verified through experiments to verify the reliability of the simulation. A micromixer with single-sided transparent materials were fabricated, and two different colors of fluids was recorded in the mixing process. As shown in Fig. 4, the two fluids diffused and mixed rapidly after contact, and the experiment and simulation in the local amplification area were in good agreement. The darker areas in the figure indicate that the concentration is high. When the fluid passes through the X-shaped structure, the diffusion and mixing between the fluids make the color lighter. After the first cubic mixing unit, the two liquids have a large contact area, which can thoroughly integrate the natural liquid. The right half is darker than the left half, which is similar to the theoretical simulation concentration distribution.

The values of the continuity equation in this paper are stabilised after a calculation step of 1000 as shown in the figure below:

From Fig. 5, the results of the continuity equation are stable at 3.7 × 10–4 and remain stable as the calculation step increases, and remain stable after reducing the residual convergence criterion, so 10–3 is chosen as the convergence criterion.

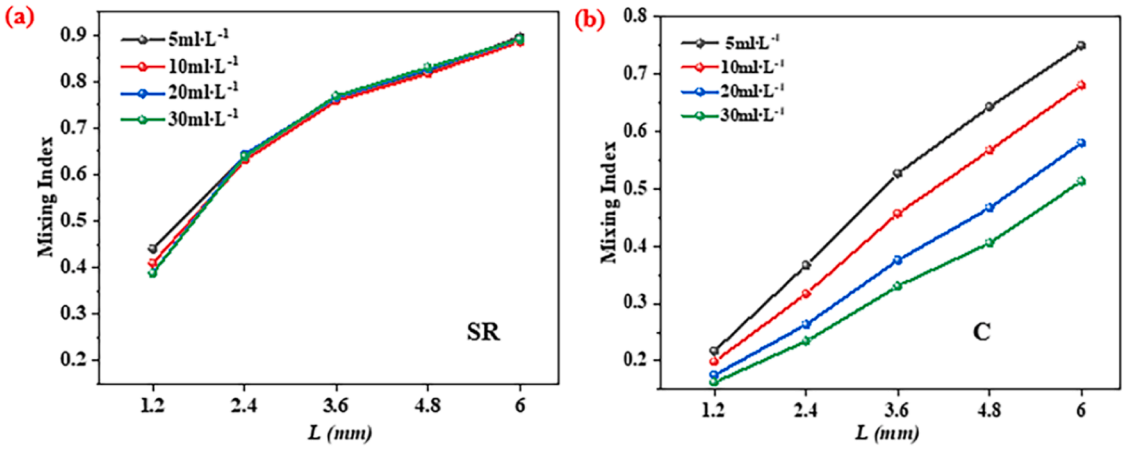



**Fig. 9.** Mixing index at different flow rates. (a) Mixed index of SR micromixer (b) Mixed index of C micromixer.

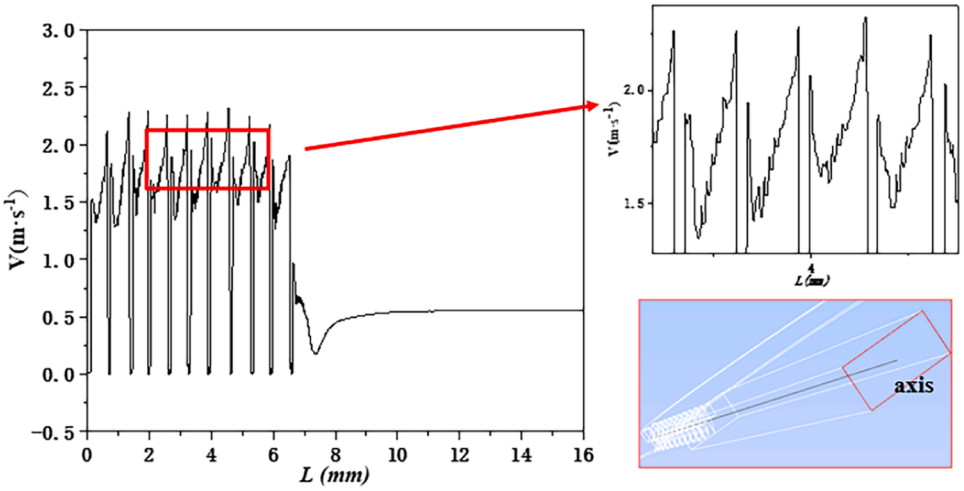



**Fig. 10.** Flow rate distribution on the central axis of SR micromixer at 5 mL • $min^{-1}$.

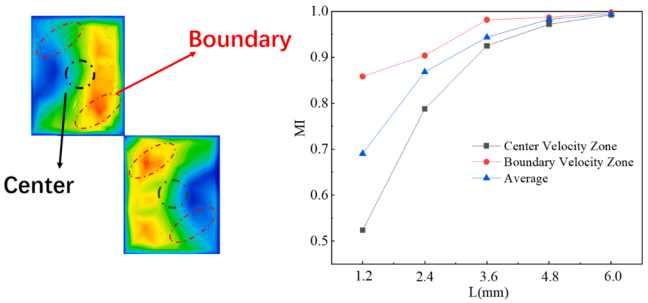



**Fig. 11.** MI at different speed zones.

### 3.2. *Influence of the mixing unit type on the mixing effect*

The mix unit structure and mixing performance of the SR micromixer and C micromixer are shown in Fig. 6. The microchannel of the C micromixer is a simple cube, realizing the free flow of fluid inside it without being affected by the structure. As shown in concentration distribution, solutions A and B began to mix violently from the first grid of the SR micromixer, while, they gradually mixed in the C micromixer. As shown in the flow rate distribution, high and low flow rates are repeated in each mixing unit in a similar distribution in SR micromixers, coexisting of large area high flow rate regions (~2.5 m/s) and small zero flow rate regions. However, flow rates are uniformly distributed at the C micromixer beside the boundary layer. The mixing indexes were quantitatively measured as the mixing index of the two micromixers. The mixing index=0 and 1 indicate that the two solutions are entirely unmixed and fully, respectively. As shown in Fig. 6, the mixing index of the SR micromixer was much higher than C micromixer at the same feed flow rate (5 mL·$min^{-1}$). The results showed that the micromixer channel design is crucial for mixing effect.

Fig. 7 shows the mixing process of SR type and C type micromixers when the two inlet flows are 5 mL·$min^{-1}$, 10 mL·$min^{-1}$, 20 mL·$min^{-1}$and 30 mL·$min^{-1}$, respectively. For the SR mixer, the two fluids are mass transferred and mixed in the mixing unit, and the fluid enters the next mixing unit and is divided into two parts, which are mass transferred and combined with the other liquid, respectively. After six mixing units, it showed uniform concentration distribution, indicating that the liquid has reached the preliminary mixing. The extended outlet pipe can realize the fluid remixing, so the concentration at the micromixer outlet is about 0.5.

The concentration distribution at different flow rates for each cross-section of the SR micromixer and C micromixer mixing structures is explored in Fig. 8. The multiple separation and recombination of the fluids in the mixing form makes the convective mixing process superior to molecular diffusion, resulting in a more uniform concentration distribution at the 6 mm cross-section. For comparison, it can be mixed entirely uniformly at a low flow rate, but can only be partially mixed at a high flow rate for C micromixers at the outlet, indicating residence time

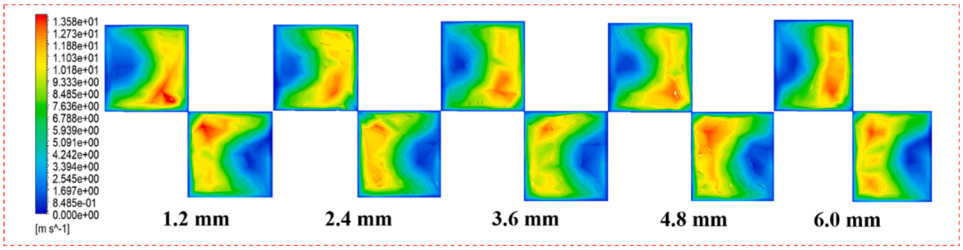



**Fig. 12.** Cloud map of velocity distribution at different cross sections.

is critical to the mixing effect for C-type micromixers. Fig. 9

Fig. 10 shows the flow rate distribution on the central axis of SR type micromixer. The flow rate variation in the composite structure with a length of 6 mm is rapid and dramatic, and the pulsation phenomenon is evident in the insert Fig. 10. The violent flow rate variation strengthens the disturbance and convection intensity between fluids, which is conducive to improving the mixing index [31]. The speed on the centerline of the extended outlet pipe is smooth, and there is no fluctuation, which is similar to C type micromixer.

Considering the effect of local velocity on MI and calculating the MI for different velocity convergence, the results are shown in the Figs. 11 and 12:

As shown in the Fig. 11, the MI at the lower boundary and centre of the different cross-sections follows the same trend as the average MI we obtained. Fig. 12 shows a cloud of velocity distributions at different cross-sections, which are similar for different cross-sections.

### *3.3. Simulation calculation for mixing unit type design*

According to the numerical simulation results of SR type micromixers and C type micromixers, the concave-convex structure of the micromixer has an essential influence on the mixing index. As a result, the internal system and size of the micromixer were designed to improve the mixing effect further. As shown in Fig. 13, XSR0.25 is an X-type fully bonded system, and the height of section B is 0.25 mm. The system of SR0.25, SR0.375, and SR0.5 were partially related, and the flowers of section B are 0.25 mm, 0.375 mm, and 0.5 mm respectively. The length of each mixing unit is 0.6 mm, and there are 10 mixing units for each micromixer. By changing the height of section B, the differences between the flow and mixing processes could be explored.

The concentration distribution diagram of XSR0.25, SR0.25, SR0.375 and SR0.5 type micromixers were compared with C type micromixers in each section as shown in Fig. 14. In the concentration distribution graph, it proved that the concentration distribution on the final section is quite different due to different mixing unit structures. The two fluids in C type micromixer present an axisymmetric distribution in the microchannel, and the concentration distribution indicated that it does not realize complete mixing at the 6 mm section. And SR0.25, SR0.375, and SR0.5 realize convection mixing due to the mixing unit structure at the 6 mm section. And it also can be concluded that the height of section B is inversely related to the mixing effect. Moreover, XSR0.25 has the same concave section height as SR 0.25, but its mixing effect t is worse than that of SR 0.25. Because the designed structure is completely bonded, and the fluid cannot be separated, the two fluids impact each other in the X-shaped system.

Fig. 15 shows the cross-sectional velocity vector of the micro-reactor. From the diagram you can infer the motion of the fluid, including rotation and binding. From the velocity vector of each section, it can be found that the fluid moves from the center of the section to the two sides, and the direction of motion reverses to form the vortex structure. The convection effect is stronger in SR0.25, and the secondary flow effect caused by the vortex structure makes the fluid mix uniformly. In SR0.375 and SR0.5, the fluid reversal trend gradually weakened, the vortex structure gradually disappeared, and the mixing effect of fluid in the channel decreased.

The mixing index of each cross-section of XSR, SR, and C-type of micromixers was quantitatively calculated in Fig. 16(a). The mixing index of SR0.25 at a 3.6 mm section exceeds 0.9, and unit can reach 0.99 at a 6 mm section. This is because the short area and sharp structure strengthen the secondary flow effect and make the fluid more uniform. [32] The mixing effect of other micromixers is reduced to a certain extent, no matter the height of the section or X-type fully bonded structure. The fluid mixing index at the outlet of the micromixer(6 mm) was calculated in Fig. 16(b). The fluid in the C-type micromixer is laminar ($Re$<1200), whose diffusion mechanism is molecular diffusion. As a result, the mixing index is positively related to the residence time and negatively associated with the flow rate. The concave-convex structure inside the XSR and SR micromixer increases the turbulence fluid effectively, and the mixing index has been significantly improved. For the XSR0.25 micromixer, the mixing effect has no significant relationship with the fluid flow rate. For XSR0.25, SR0.375, and SR0.5 micromixer, the mixing coefficients decreased to some extent, and the mixing index was first decreased and then increased. This shows that molecular diffusion is shown at a low flow rate (<10 mL·min$^{-1}$), and

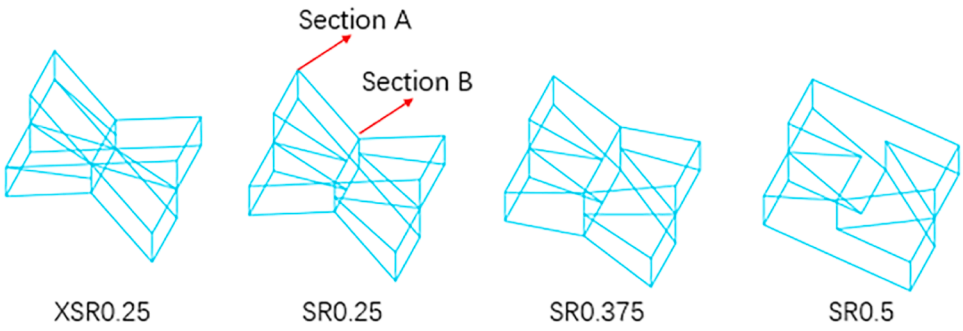



**Fig. 13.** The internal structure of designed micromixer.

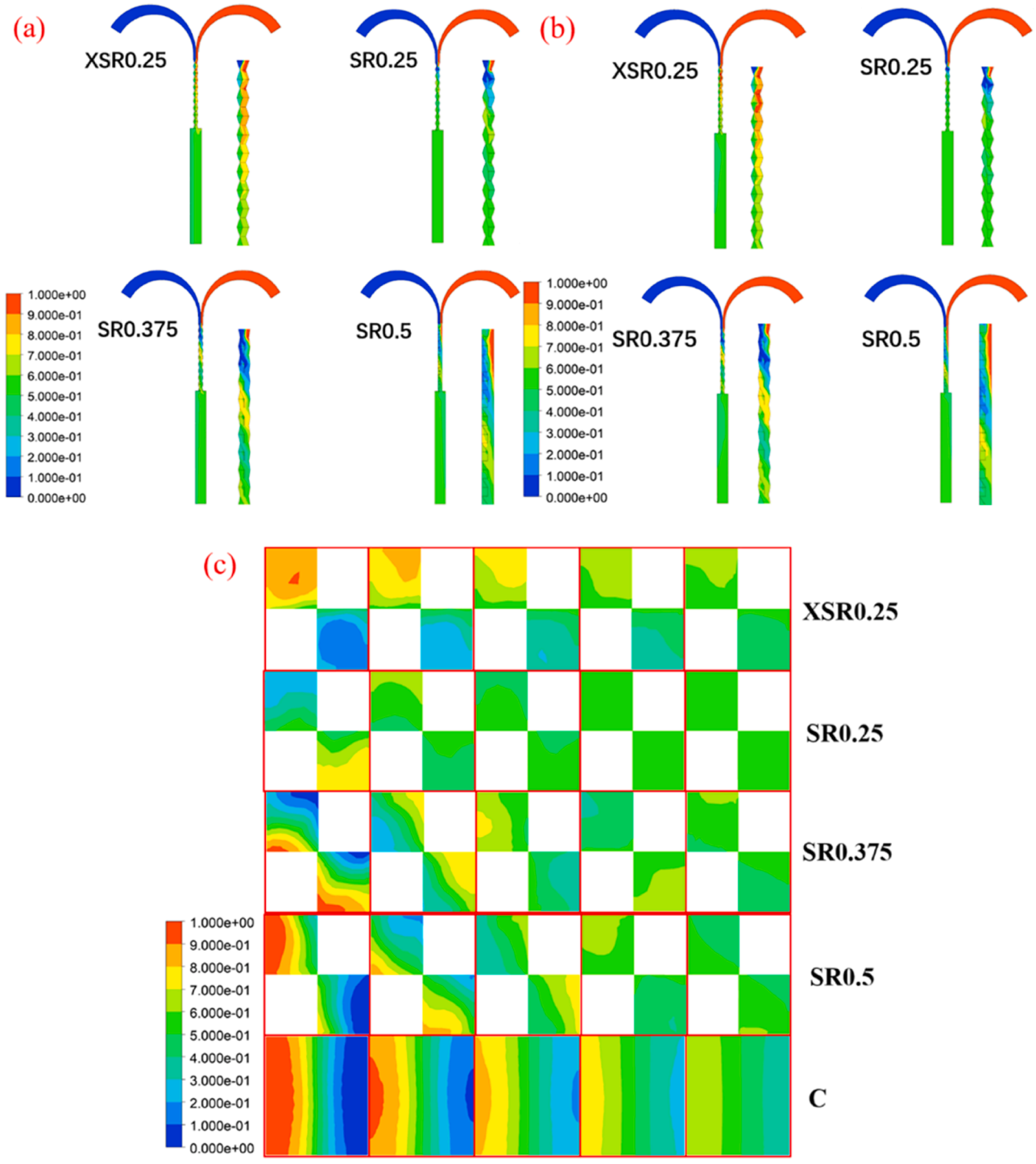



**Fig. 14.** Concentration distribution of XSR0.25, SR0.25, SR0.375 and SR0.5 micromixers at (a)5 mL·min$^{-1}$ and (b)20 mL·min$^{-1}$. (c) Concentration distribution of XSR0.25, SR0.25, SR0.375, SR0.5 and CSR micromixers in each section at 5 mL·min$^{-1}$.

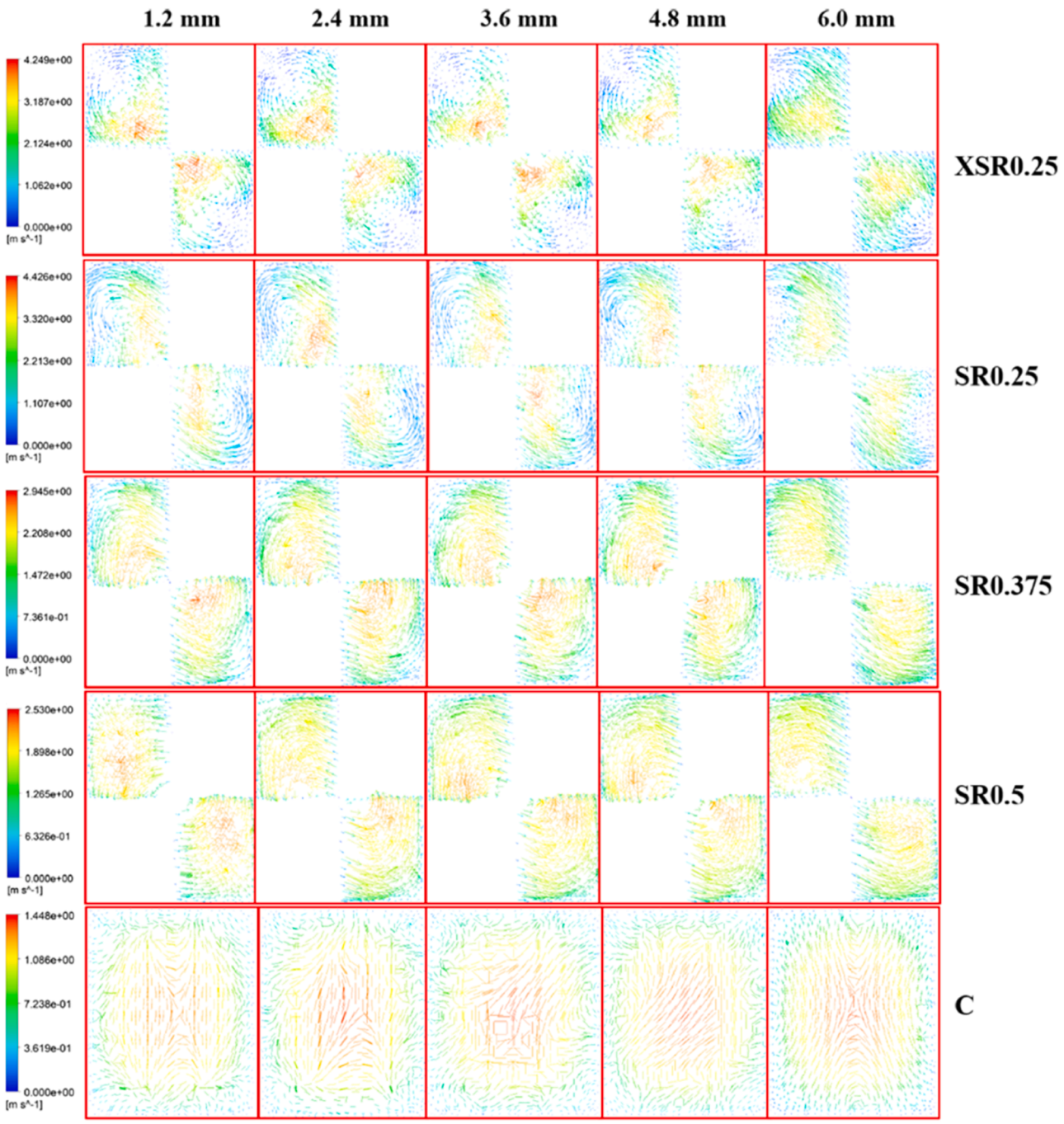



**Fig. 15.** Velocity vector diagram of XSR0.25, SR0.25, SR0.375, SR0.5, and C reactors in each section at 5 mL·min$^{-1}$.

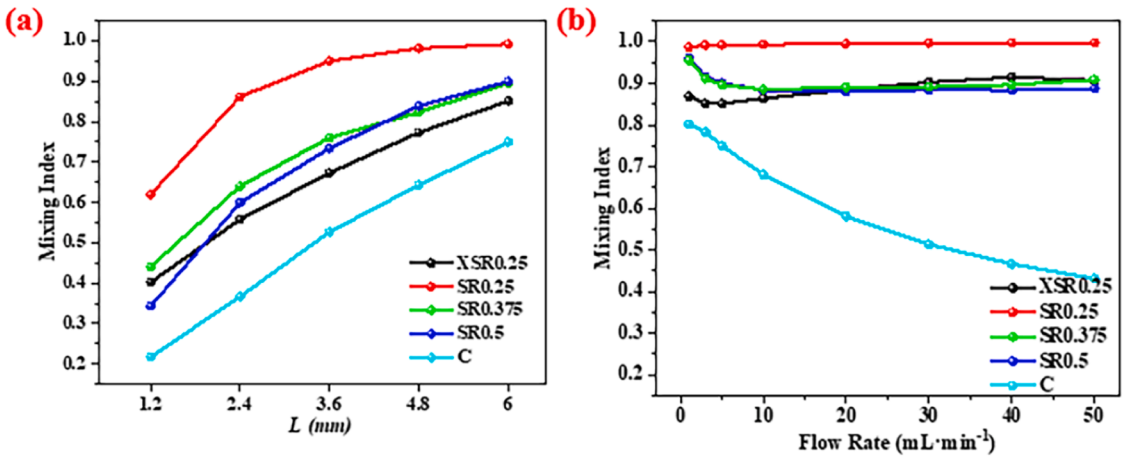



**Fig. 16.** (a) The mixing index of XSR0.25, SR0.25, SR0.375, SR0.5, and C micromixers in each section at 5 mL·min$^{-1}$ (b) Effect of Flow Rate on the mixing index at outlet of micromixer.

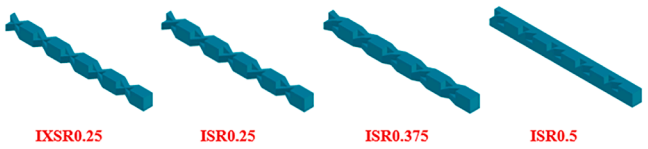



**Fig. 17.** Recombination structure of micromixer.

convection-diffusion is shown at a high flow rate (>10 mL·min$^{-1}$). It can be concluded that the concave-convex structure inside the micromixer decreases the critical flow rate from laminar flow to turbulent flow.

### *3.4. Simulation calculation for structural recombination*

The structure was reorganized and designed by structural recombination to reduce the complexity of the micromixer and reduce the flow resistance as well as ensue the mixing effect performance, as shown in Fig. 17. The recombination unit of each model is combined with a cube, for instance an IXSR0.25 team is composed of an XSR0.25 unit with 0.6 mm length and a cube unit with 0.6 mm size. The length of the recombination structure micromixer is the same as that of the original micromixer, and each micromixer contains 5 mixing units.

Fig. 18 displays the concentration distribution in the recombination structure at flow rates of 5 mL·min$^{-1}$, respectively. The concentration at the outlet of the composite form is concentrated at about 0.5 after recombination, indicating a completely mixed of two fluids. Compared with XSR0.25, SR0.25, SR0.375, and SR0.5 micromixer, the mass transfer process in IXSR0.25, ISR0.25, ISR0.375, and ISR0.5 of fluid separation and recombination is weakened. Surprisingly, the mixing effect did not decrease due to the recombination of cubic structure, but all improved. The reasons were speculated that the cube structure effectively increases the contact area between fluids, which increases the mixing efficiency.

The mixing index at the outlet of the mixing structure under different flow rates of each model was calculated to verify the mixing effect of the micromixer at 1–50 mL • min$^{-1}$ in Fig. 19. The mixed index of ISR 0.25 remained above 0.99, and the diverse index of ISR 0.375 and ISR 0.5 were improved. The IXSR0.25 mixing index rapidly increases to above 0.9 with a high flow rate (>3 mL • min$^{-1}$), which is due to the flow rate increase strengthening the secondary flow effect in the cubic structure [33]. In particular, IXSR0.25 mixing performance is similar to SR0.25 when the flow rate is 50 mL • min$^{-1}$, and this structure is suitable for fluid mixing at high flow rates.

## 4. Conclusions

A numerical simulation is characterized by the flow regimes appearing in SR, allowing quantitative comparison to the C micromixer. Based on diffusion in C mixer, SR can achieve split and recombine multiple times. Concurrently, the results show that the chaotic convection and secondary flow effect is weakened, and the mixing worsens under the high depth section. In contrast, SR0.25 achieved good mixing performance and the mixing index reached 0.99 within the scope from 1 mL min$^{-1}$ to 50 mL min$^{-1}$, which is mainly due to chaotic convection. In allusion to SR0.375, SR0.5 and XSR0.25 carrying out slightly mixing, the proposed micromixers equipped with the cubic structure are designed to enhance the mixing performance. This is because the system can provide a large contact area and generate the stronger Dean Vortex with increasing flow rate. The micromixers are designed that make full use of a relatively short channel, which is optimized to improve the mixing intensity of the micro-reaction system at a range from 1 mL min$^{-1}$ to 50 mL min$^{-1}$. Therefore, these micromixers will have a broad application prospect.

A comprehensive analysis of the mixing process of each model shows that convective mass transfer and molecular diffusion mass transfer play an essential role in completing the initial mixing of the fluid, and the more fully mixed liquid is due to the secondary flow effect in the mixing structure. This chapter provides a research basis for the design of the micromixer, and the mixing result of the improved model is greatly improved. This design idea has laid a particular research foundation for the developing of the SAR micromixer.

## 5. Table of symbols

| Type | Full name |
|---|---|
| **C** | cube micromixers |
| **SR** | Split and Recombine |
| **XSR** | X shape- Split and Recombine |
| **IXSR** | recombination X-shaped split and recombine micromixer |

## CRediT authorship contribution statement

**Lei Zhang:** Conceptualization, Investigation, Writing – original draft, Visualization. **Jia Li:** Investigation. **Mingxing Liu:** Investigation. **Guiying Tian:** Conceptualization, Supervision, Writing – review & editing. **Penggao Cheng:** . **Jianping Zhang:** . **Na Tang:** Conceptualization, Supervision, Funding acquisition.

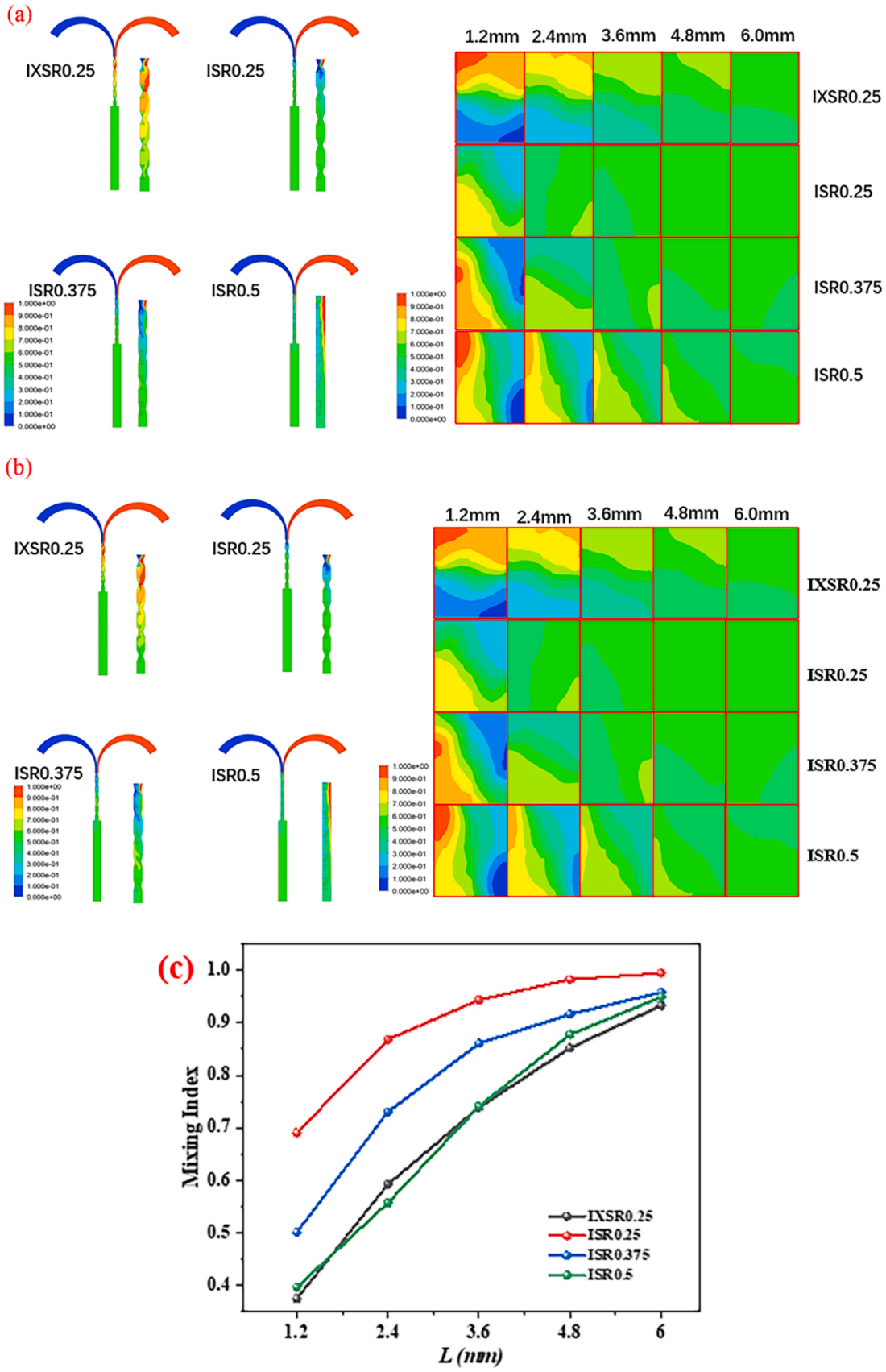



**Fig. 18.** Concentration distribution of IXSR0.25, ISR0.25, ISR0.375 and ISR0.5 micromixers at (a) 5mL·min$^{-1}$ (b) 20 mL·min$^{-1}$ (c) The mixing index of IXSR0.25, ISR0.25, ISR0.375 and ISR0.5 reactors in each section.

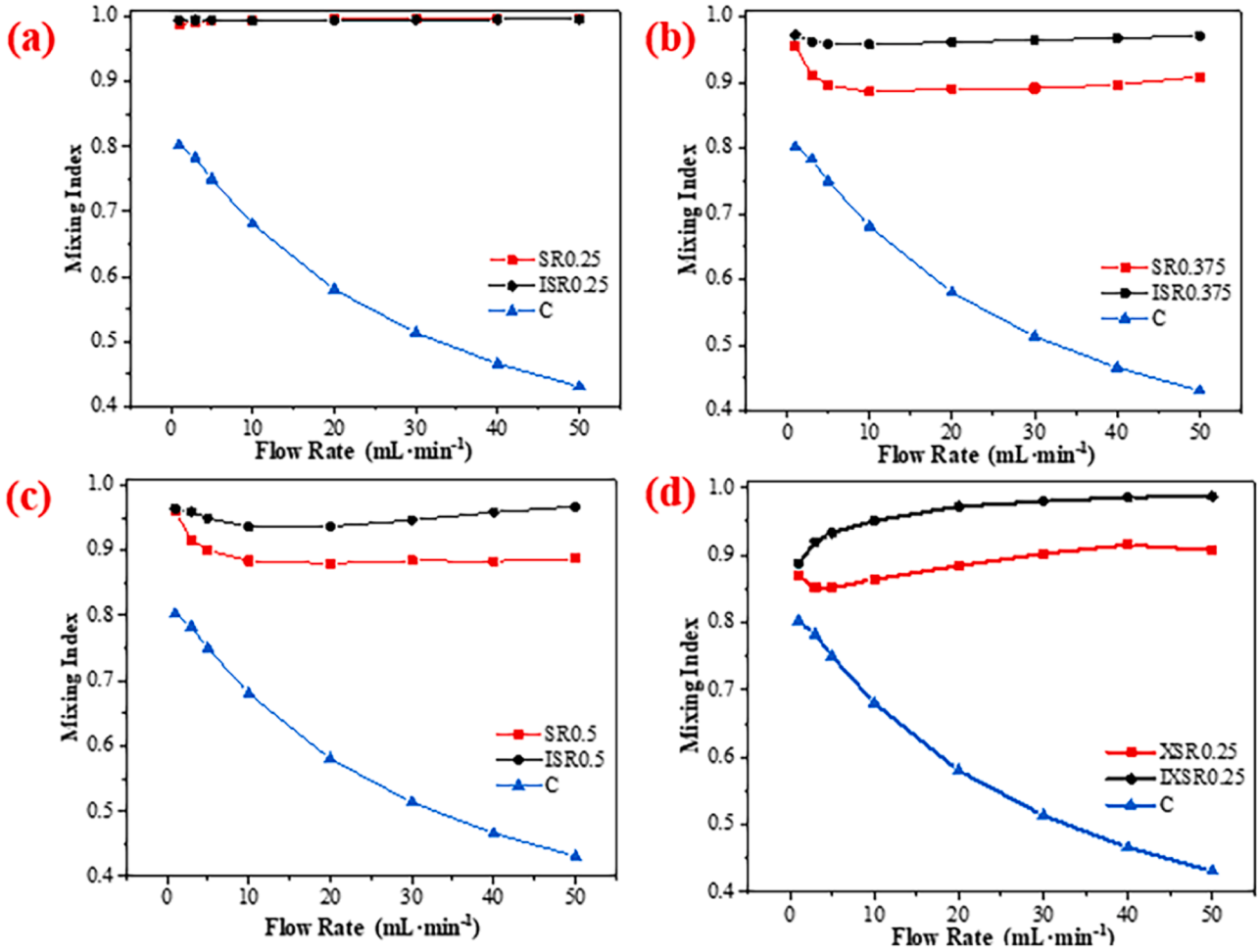



**Fig. 19.** Effect of flow rate on the mixing index.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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